

Executive Summary

Open Neighbourhoods Precede Concept Spread: Structural Antecedents and Cross-Disciplinary Diffusion of Emerging Scientific Concepts

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Headline Finding

When a scientific concept enters a new disciplinary subfield, its subsequent uptake by that subfield’s own newcomer authors is predicted by how host-native its initial co-occurrence partners are. The MeSH biomedical replication gives an incidence-rate ratio per standard deviation of host-nativeness (IRR/SD) of **1.23 [1.12, 1.36]** (Holm $p < 0.001$). The held-out confirmation on the main arXiv-sourced pool gives $\text{IRR/SD} = 1.19 [1.06, 1.33]$ ($p = 0.004$). The inverse-variance-weighted pooled estimate across both populations is 1.26 [1.17, 1.36] with $I^2 = 0$. Co-transfer of origin companions is null across all folds.

Study Design

The study investigates two research questions using an OpenAlex-based scholarly dataset of 426 semantically grounded concepts (366 main, 60 reference) with 462,812 works:

- **RQ1 (Emergence):** Which temporal and structural patterns in an evolving scientific knowledge graph characterise the emergence of scientific concepts?
- **RQ2 (Diffusion):** How do emerging scientific concepts diffuse across disciplinary communities, and which temporal network patterns distinguish locally concentrated from broadly integrated concepts?

The 366 main concepts are split into a 247-concept screen fold and a 119-concept held-out fold. Twenty-five yearly co-word network snapshots ($\approx 27,000$ nodes, $\approx 84,000$ edges) were built from a design-weighted background sample and partitioned into Leiden communities. An independent MeSH biomedical population of 191 concepts serves as a replication arm. All decision rules, kill rules, and specifications were pre-registered and hashed before inspection.

RQ1: Emergence — Primary Closure Dead, Persistent-Neighbour Confirmed

The original hypothesis that higher local closure precedes concept emergence was tested with a pre-registered event study. The primary closure measure did not survive held-out confirmation

(**R1_DEAD**: Holm $p = 0.147$). This outcome was governed by the frozen kill rule: the held-out confidence interval includes zero, so the primary claim is dead.

Only **persistent-neighbour closure** survives: held-out $S = -1.073$, 95% CI $[-1.858, -0.289]$, Holm $p = 0.015$. This means that before a concept shows sustained uptake, the subset of its co-occurrence neighbours that persist across yearly snapshots becomes less clustered — the concept’s stable neighbourhood opens up.

The mechanism is **not Burt brokerage**: Burt constraint is *higher*, not lower, before emergence ($S = +0.083$ [0.017, 0.147] on screen, $+0.105$ [0.021, 0.187] on held-out). Pre-emergence openness does not predict later host-entry anchoring (closure–anchoring link null: partial $\rho = 0.049$, $p = 0.564$ pooled). The emergence and diffusion findings are therefore independent.

Precursor measures add no predictive value for *whether* a concept will show sustained uptake (delta-AUC = -0.001 [$-0.008, 0.006$]).

RQ2: Diffusion — Host-Entry Grafting Confirmed and Replicated

The central finding concerns **host-entry grafting**: when a concept first appears in a new subfield, the host-nativeness of its co-occurrence partners at entry predicts how many newcomer authors in that subfield subsequently adopt it.

Table 1: Grafting results across folds and populations.

Fold	IRR/SD	95% CI	p	N
Screen (co-primary)	1.30	[1.16, 1.45]	10^{-5}	1,544
Held-out (co-primary)	1.19	[1.06, 1.33]	0.004	972
MeSH (co-primary)	1.23	[1.12, 1.36]	9.7×10^{-5}	2,171
IVW pooled	1.26	[1.17, 1.36]	—	—

Decomposition results (K-tests):

- **K1 — Both margins**: The effect operates through both the extensive margin (whether any newcomer adopts; IVW $+6.43$ pp/SD [4.76, 8.11]) and the intensive margin (how many adopt, given ≥ 1 ; IVW IRR/SD 1.183 [1.104, 1.267]). The extensive margin accounts for 38% [21%, 55%] of the total log-IRR.
- **K2 — Host-specific**: Adding generality controls removes only 3.1% of the effect (IVW retention 0.969 [0.81, 1.10]). The Balassa-index lift measure replicates across all folds (IVW IRR/SD 1.34 [1.23, 1.46]).
- **K3 — No field boundary**: The effect does not vary by origin field (Wald $p = 0.638$).

Adopter-level mechanism. Authors who adopt a concept in a new subfield are enriched for prior exposure to the concept’s entry partners (OR 3.09 [2.32, 4.28]), with the gradient running FOREIGN > ADJACENT > NATIVE. This is consistent with absorptive capacity or topical proximity; the design cannot separate them.

Diffusion Typology and Temporal Ordering

Only the $k = 2$ typology is stable (min Jaccard 0.861): **localised** ($n = 136$) and **broad from the start** ($n = 66$). Finer typologies are unstable and add nothing beyond entropy trajectories. Network expansion precedes disciplinary diffusion in 25 of 26 pooled main-arm concepts with both transitions observable.

Key Dead Ends

1. **Source-sink viability blocked.** Gate A failed: only 17.3% of host edge-years have within-host traced share ≥ 0.40 (mean 0.204). The viability decomposition could not be executed.
2. **Openness does not predict where concepts travel.** All breadth-prediction Holm $p = 0.405$.
3. **Brokerage is not the mechanism.** Burt constraint is positive, opposite to prediction.
4. **Co-transfer of origin companions is null** across all folds and populations.
5. **Precursors do not predict whether a concept emerges** beyond volume and burst baselines.

Caveats and Open Questions

- The concept pool is dominated by physics and computer science (arXiv). Social sciences and humanities are untested.
- 87% of screen co-primary events are single-paper host entries. The held-out single-paper interaction is significant ($p = 0.004$).
- Binary establishment (EST_bin) is null on the held-out fold ($p = 0.16$). The claim holds for uptake initiation but not strict establishment.
- On MeSH, 12.4% of the effect is absorbed by field breadth.
- The full numerical audit found 292 numbers present in artifact files but not transcribed into the main report, with seeded-perturbation recall of 0.60.

Target Venue

Applied Network Science.